

Software Engineering & Project Management		Semester	V
Course Code	BCS501	CIE Marks	50
Teaching Hours/Week (L:T:P: S)	4:0:0:0	SEE Marks	50
Total Hours of Pedagogy	52 hours	Total Marks	100
Credits	04	Exam Hours	03
Examination nature (SEE)	Theory		

Passing Package or (Important Questions)

Module 1:

- 1) Define Software Engineering and explain practices involved in software Engineering.
- 2) Define the software application domains and characteristics of software.
- 3) Explain unique nature of Web Apps.
- 4) Explain seven principles that focus on software engineering.
- 5) Explain new Software Challenges and Software Myths.
- 6) Demonstrate Waterfall, V- Model with a neat, labelled diagram.
- 7) With a neat diagram explain Incremental Process Models.
- 8) Briefly explain the Spiral Model with neat diagrams.
- 9) Explain The Unified Process.

Module 2:

- 1) Explain how Requirement Engineering works.
- 2) Negotiating requirements & Validating requirements.
- 3) Explain key steps are involved in establishing the foundational groundwork for a successful software engineering project.
- 4) Provide a brief overview of the key tasks involved in requirement engineering.
- 5) Demonstrate access camera surveillance via Internet – display camera views function with a neat activity diagram.
- 6) Explain key steps involved in building a requirement model, and how do they contribute to the software development process?
- 7) Demonstrate class-based modelling and identify the analysis classes.
- 8) Develop UML usecase and activity diagrams.
- 9) Explain Flow oriented Modeling, Behavioral Modeling.

Module 3:

- 1) Explain Agility, Agile Process and its principles.
- 2) Demonstrate Extreme Programming Process with a neat, labelled diagram.
- 3) Write short notes on tool set for Agile Process.
- 4) Explain the principles that guide each framework activity in the software engineering process.
- 5) Describe software engineering knowledge and its core principles.
- 6) Write short notes on other Agile software process.
- 7) Explain Scrum, Crystal with diagram.
- 8) Explain ASD and FTD Model with diagram.

Module 4:

- 1) Describe Project Management life cycle and Sequence of Development activities.
- 2) Demonstrate activities covered by software project management with neat diagram.
- 3) Explain various stages of contract management.
- 4) Explain project risk evaluation.
- 5) Explain cost benefits of evaluation techniques.
- 6) Differentiate between Software Projects V/S Other types of projects.
- 7) How do we categorize the projects under software project management.
- 8) Short note on Project Success and Failure, SMART objectives .

Module 5:

- 1) Define Software Quality and explain the place of software quality in project management and importance of software quality.
- 2) Explain Software Quality Models.
- 3) Differentiate between Product and Process Quality Management.
- 4) Describe decomposition techniques in software project estimation.
- 5) Explain Empirical Estimation model.
- 6) Write a short note on Observations on project estimations.